

Project: **Amis to raise awareness about Neutropenia fever and implement a proactive monitoring system for patients at high risk**

Program Duration

10th September 2024 – 15th March 2025

Program Objective

The primary objectives of this project are:

- **Raise Awareness:** Educate patients and their families about the risks of neutropenia fever and the importance of early detection and timely medical intervention.
- **Provide Monitoring Devices:** Distribute continuous temperature monitoring devices to patients who have undergone cancer treatment and chemotherapy, focusing on those at high risk for neutropenia fever.
- **Enable Early Detection:** Utilize technology to detect fever symptoms early, allowing for quick alerts to healthcare providers and prompt hospital admissions if necessary.
- **Impact Patient Lives:** Patients to receive the benefits of early fever detection and intervention.

INTRODUCTION

Status of Cancer in Bihar

(based on secondary literature review, interaction with tertiary health centres and specialised cancer hospitals in the state)

Cancer in Bihar represents a growing public health concern, accounting for a significant proportion of India's cancer burden. According to data from the National Cancer Registry Program (NCRP), the state has seen a rise in cancer cases, particularly those linked to tobacco use and infections. Bihar's cancer incidence rate is estimated at 76.4 cases per 100,000 population annually, which is slightly below the national average but reflective of underreporting in rural regions.

Prevalence and Incidence

Cancer Types:

- Men: Tobacco-related cancers such as oral, esophageal, and lung cancers constitute over 40% of cases in men.
- Women: Breast cancer (20%) and cervical cancer (12%) are the leading causes of cancer morbidity.
- Approximately 50% of all cancer cases in men and 15% in women are linked to tobacco use. Bihar has one of the highest rates of smokeless tobacco use in India, with 53.5% of adult males and 6.8% of females consuming smokeless forms.
- Paediatric cancers account for 4-5% of all cancer cases in Bihar, primarily leukemia and lymphomas. Late diagnosis remains a critical issue.

Neutropenic fever, characterized by a fever in individuals with abnormally low levels of neutrophils (a type of white blood cell), is commonly observed in patients undergoing chemotherapy or those with conditions affecting bone marrow function. Specific data on the incidence or prevalence of neutropenic fever in Bihar is not readily available in public health records. This lack of data may be due to underreporting or the condition being managed within specialized healthcare settings without extensive public documentation.

However, Bihar has been confronting outbreaks of other febrile illnesses in recent years:

- **Acute Encephalitis Syndrome (AES):** In June 2019, Bihar experienced a significant outbreak of AES, colloquially referred to as 'brain fever.' The state health department reported over 600 cases across 16 districts, resulting in 136 fatalities. Muzaffarpur district was notably affected, accounting for 117 of these deaths. Experts linked the high incidence of hypoglycemia (low blood sugar) in malnourished children to the consumption of unripe litchi fruit, which contains a toxin that can precipitate this condition.
- **Dengue Fever:** The state has also been grappling with dengue outbreaks. In November 2023, Bihar registered 197 new dengue cases in a single day, bringing the total number of cases that year to 18,601. Patna reported the highest number of new cases, followed by Bhagalpur, Gaya, Vaishali, and Nalanda.
- **Scrub Typhus:** Amid the ongoing dengue outbreak, Bihar witnessed a surge in scrub typhus cases in October 2023. This bacterial infection, transmitted through bites of infected chiggers, presented diagnostic challenges due to symptom overlap with other febrile illnesses. Delayed or incorrect diagnoses led to fatalities and complications such as multi-organ dysfunction.

Neutropenic fever, a serious complication often arising during chemotherapy-induced neutropenia, poses significant challenges in rural settings like the villages of Bihar.

SURVEY

A survey of 50 caregivers across South Bihar (Gaya, Jamui, Aurangabad, Patna) was conducted, supplemented with a review of existing literature and regional data on chemotherapy complications. Caregivers included spouses, parents, children, and other immediate family members.

Key Findings:

1. Relationship with the Patient:

- 60% caregivers were immediate family (spouse or parent).
- 30% were adult children.
- 10% were other relatives or friends.

2. Major Challenges Faced:

- **Emotional Stress:** 80% of caregivers reported mental exhaustion and anxiety.
- **Physical Fatigue:** 70% cited exhaustion from caregiving duties.
- **Economic Burden:** 50% struggled with financial constraints due to medical costs and loss of income.

3. Complications During Chemotherapy:

- 90% noted nausea and vomiting in patients.
- 50% reported depression and emotional instability.
- **Neutropenia Fever Awareness:** Only 20% of caregivers were aware of the risks and signs of neutropenia fever, despite its severe implications.

4. Neutropenia Fever: A Silent Threat

- Neutropenia fever, caused by a dangerously low white blood cell count, makes patients highly susceptible to infections.
- According to the **Journal of Global Oncology**, 48% of cancer patients in India develop neutropenia during treatment, with nearly 20% resulting in febrile neutropenia.
- Caregivers lacked resources and awareness for early detection, leading to delays in medical intervention.

5. Coping Mechanisms:

- 60% consulted doctors to manage side effects.
- 30% used traditional remedies or community advice.
- 10% sought professional counseling.

6. Support Systems:

- Only 40% felt adequately supported by healthcare providers.
- 30% depended on personal networks for assistance.

- 30% expressed dissatisfaction with the support mechanisms.
- 7. **Impact on Daily Life:**
 - 50% experienced disruptions in professional responsibilities.
 - 40% reported neglect of other familial obligations.
- 8. **Demand for Additional Support:**
 - **Financial aid** (70%).
 - **Access to trained healthcare workers** (50%).
 - **Monitoring tools for complications like neutropenia** (60%).

Observations:

- Neutropenia-related complications pose a significant, under-addressed risk for chemotherapy patients in South Bihar.
- Lack of awareness and monitoring of neutropenia fever increases morbidity and mortality risks.

Caregivers face a dual burden of emotional stress and insufficient resources for managing severe complications like febrile neutropenia.

PROGRAM LAUNCH: Event at Patna



Date: 30th November 2024

The Neutropenia Fever Awareness Program was successfully conducted in Patna, Bihar, with the objective of raising awareness and enhancing knowledge about neutropenia fever. The event aimed to educate healthcare professionals, biomedical engineers, and stakeholders on the causes, symptoms, prevention, and treatment of this critical health issue. Organized by [Insert Organizing Body/Institution], the event witnessed the participation of distinguished medical professionals, biomedical engineers, and healthcare stakeholders.

The event featured prominent speakers, including Dr. Soni Rani (MBBS, MD, Assistant Professor, RDJMMCH, Muzaffarpur), Dr. Arun Kumar Pandey (MBBS, MD, PGDE, Ex-SMO WHO India, Associate Professor & Head, RDJMMCH, Muzaffarpur), Kanika Bansal (Biomedical Engineer), and Sunny Jain (NICS representative). Their collective expertise provided a comprehensive understanding of the complexities surrounding neutropenia fever.

The event commenced with an inaugural session where the organizing committee welcomed attendees and emphasized the need to address the rising cases of neutropenia fever. This session set the tone for the day's proceedings, highlighting the relevance of timely diagnosis and treatment to reduce mortality rates, especially in immunocompromised patients undergoing chemotherapy.

The expert lecture series began with Dr. Soni Rani, who provided a detailed overview of the pathophysiology of neutropenia fever. She stressed the need for early detection and timely intervention. Her lecture highlighted the challenges posed by the disease, especially for cancer patients, and the critical role healthcare workers play in mitigating its impact.



Dr. Arun Kumar Pandey took the stage next, presenting insights on clinical management and treatment protocols for neutropenia fever. Drawing on his extensive experience with WHO and as Head of RDJMMCH, he emphasized the importance of adhering to standard operating procedures (SOPs) and following antibiotic stewardship. His insights on infection control and the role of paramedical staff in managing febrile neutropenia were particularly impactful.

The technical session saw Kanika Bansal, a Biomedical Engineer, demonstrating the tool **NV-Core**, an advanced device used for measuring body temperature. This device, with its precise and real-time measurement capability, is designed to support the early detection of febrile neutropenia. Attendees were given a live demonstration of how the NV-Core tool works, with healthcare providers and biomedical engineers actively engaging

in the session. Kanika Bansal emphasized how technological innovations like the NV-Core could revolutionize infection control and improve patient outcomes, especially in resource-limited settings.

Sunny Jain from NICSI showcased the role of digital technology in healthcare, discussing NICSI's initiatives to support hospitals and healthcare providers. His presentation highlighted the use of real-time patient monitoring tools, which aid in the early detection of complications in febrile neutropenia patients. The integration of these digital tools is expected to enhance healthcare delivery in remote and underserved areas.

The event also included an interactive panel discussion where experts addressed questions from attendees. Key discussion points focused on the challenges of diagnosing neutropenia fever in rural healthcare settings, the role of nurses in patient care, and the integration of new biomedical devices in hospitals. The session provided an open forum for dialogue between healthcare providers, biomedical engineers, and technology developers.



A significant highlight of the event was the hands-on training and live demonstration of biomedical devices, particularly the NV-Core temperature measurement tool. Doctors, nurses, and healthcare workers actively participated in this session, learning how to effectively use point-of-care testing (POCT) devices to identify febrile neutropenia cases early. This training equipped participants with practical skills to improve diagnosis and

treatment outcomes. The event concluded with a vote of thanks from the organizing committee. The speakers and attendees were lauded for their active participation, and special gratitude was extended to RDJMMCH Muzaffarpur for its support. Attendees appreciated the event's focus on practical training and the introduction of technological tools like NV-Core.

SENSITIZATION & AWARENESS WITH HOSPITALS, CAREGIVERS & FRONTLINE HEALTH WORKERS



Challenges in Managing Neutropenic Fever in Rural Bihar

Managing neutropenic fever in rural Bihar presents unique challenges due to infrastructure limitations, socio-economic barriers, and lack of awareness. Key challenges include:

1. Limited Healthcare Infrastructure

- **Diagnostic Gaps:** Rural healthcare centers often lack advanced diagnostic tools such as blood culture systems, complete blood count (CBC) machines, and biomarker tests (like procalcitonin) critical for neutropenic fever diagnosis.
- **Lack of Specialized Care:** Neutropenic fever often requires intensive care, which is scarce in remote areas.
- **Delayed Referrals:** Patients may be referred to tertiary hospitals only when their condition worsens, reducing the chances of successful treatment.

Mitigation Strategies:

- **Upgrade Primary Healthcare Centers (PHCs):** Equip PHCs with basic diagnostic tools and train staff to recognize and manage early-stage febrile episodes.
- **Telemedicine Support:** Establish teleconsultation links between village health centers and specialized hospitals to enable expert guidance on critical cases.
- **Fast-Track Referral Systems:** Develop streamlined referral protocols to ensure timely patient transfers to equipped facilities.

2. Poor Access to Medicines and Treatment

- **Availability Issues:** Essential medications like broad-spectrum antibiotics, antifungals, and growth factors (e.g., G-CSF) are often unavailable.
- **Cost Constraints:** Even if available, these medications may be unaffordable for low-income families.

Mitigation Strategies:

- **Medicine Banks at PHCs:** Maintain dedicated stockpiles of essential drugs in rural clinics to support urgent treatment.
- **Government Subsidy Programs:** Implement subsidized treatment plans specifically for cancer patients and immunocompromised individuals.

- **Community-Based Insurance Models:** Encourage the adoption of low-cost insurance schemes to help families bear medical expenses.

3. Limited Awareness and Delayed Presentation

- **Lack of Awareness:** Rural communities may not recognize neutropenic fever as a life-threatening condition, leading to delayed treatment.
- **Home Remedies and Traditional Healing Practices:** Preference for traditional medicine may delay effective treatment.

Mitigation Strategies:

- **Health Education Campaigns:** Organize village-level awareness drives focused on febrile conditions in cancer patients or those with weakened immunity.
- **Training Local Healthcare Workers (ASHA/ANM):** Equip community health workers with basic knowledge to identify and respond to neutropenic fever symptoms.

4. Hygiene and Sanitation Challenges

- **Infection Risks:** Poor hygiene practices and contaminated water sources increase infection risks, which is critical for immunocompromised patients.
- **Limited Isolation Facilities:** Neutropenic patients require isolated care to prevent secondary infections, which is difficult in crowded households or rural hospitals.

Mitigation Strategies:

- **Home-Based Infection Control Training:** Educate caregivers on maintaining hygiene for vulnerable patients at home.
- **Dedicated Isolation Units in PHCs:** Establish basic infection-controlled areas in rural healthcare centers for patients at higher risk.

5. Data and Surveillance Gaps

- **Limited Case Tracking:** Neutropenic fever cases are often underreported, making it difficult to track trends or respond to outbreaks.
- **Lack of Research:** Limited research on regional microbial patterns makes empirical treatment less effective.

Mitigation Strategies:

- **Data Collection Mechanism:** Implement electronic medical records at rural centers to better track and analyze neutropenic fever trends.
- **Partnership with Research Institutions:** Collaborate with medical colleges and research bodies to gather data and develop tailored treatment protocols.

6. Nutritional and Lifestyle Challenges

- **Malnutrition in Vulnerable Groups:** Malnourished individuals, particularly children, are at higher risk of severe infections during neutropenia.
- **Environmental Risk Factors:** Poor air quality, unsafe water, and inadequate housing conditions can worsen outcomes.

Mitigation Strategies:

- **Nutritional Support Programs:** Integrate nutrition-focused interventions for vulnerable groups, especially those undergoing chemotherapy or immunosuppressive treatments.
- **Water and Sanitation Improvements:** Promote water purification techniques and improve sanitation infrastructure in rural households.

KEY INSIGHTS FROM HEALTHCARE PROFESSIONALS

- Doctors appreciate the real-time monitoring system but face challenges due to limited emergency care facilities in Jamui.
- Many cancer patients arrive at hospitals only in late stages, highlighting low awareness and detection rates.
- Due to lack of oncology departments in district hospitals, patients are often referred to Patna, Ranchi, or Kolkata, leading to treatment discontinuity.



Lack of Local Healthcare Infrastructure

- No specialized cancer treatment units in district hospitals, making referrals mandatory for severe cases.
- Limited ICU facilities hinder emergency management of cases.

Resistance to Technology Adoption

- Initial hesitation from rural caregivers in trusting and using digital monitoring devices.
- Requires continuous handholding and awareness for sustained adoption.

Financial and Logistical Barriers

- Many patients travel out of state for treatment, making continuous monitoring difficult.
- High out-of-pocket expenses for travel and treatment discourage follow-ups.

MONITORING & INTERACTION WITH MEDICAL CARE FACILITIES

- Real-Time Data Collection & Assessment:

25 continuous temperature monitoring devices were placed at designated centers to collect real-time patient data and evaluate program feasibility.

- Operational Impact on Healthcare Workers:

Healthcare staff reported improved accessibility and efficiency in patient monitoring.

Continuous temperature tracking enabled early detection of fever episodes.

- Doctor's Feedback on Adaptation:

Doctors acknowledged the necessity of real-time temperature monitoring for cancer patients but advised a cautious approach to direct implementation.

They recommended broader trials before full-scale adoption in oncology settings.

- Expanded Use as Recommended:

In response to recommendations, devices were deployed for typhoid and fever patients, including a few pediatric cases.



This helped demonstrate the effectiveness of continuous monitoring in diverse patient categories.

- Hospital Adoption Challenges & Recommendations:

A phased integration approach with expanded trials across multiple conditions was recommended to establish credibility and long-term hospital adoption.

We have deployed 25 devices on field. We have been able to reach out to 200 patients/users till 15th March 2025. We expect the numbers to increase and technology have a snowball effect with the institutions.

Month	Training Awareness & Sessions Conducted	Devices Deployed (Cumulative)	Patients Monitored (Cumulative)	Fever Incidents Detected	Medical Interventions	Key Observation
October	Team Training	-	-	-	-	Initial training phase;
November	Awareness sessions in Jamui & Aurangabad & Patna	-	-	-	-	Increased engagement, understanding challenges on field
December	Community awareness drives, interactive Q&A Session on neutropenia fever & early detection	-	-	-	-	Sensitization about nebrile fever is very limited. With overwhelmed infrastructure it's a very niche field for exploration
January	Session on neutropenia fever & early detection	5	12	2	1	Better understanding of alert protocols
February	Session on neutropenia fever & early detection	10	75	7	5	People understanding the importance of timely monitoring. Expansion beyond nebrile fever
March	Session on neutropenia fever & early detection	10	200	14	7	Out of 14 high-risk fever alerts, 7 patients received timely treatment, reducing potential complications.

						Increased Awareness: Awareness sessions have helped over 500 individuals (cumulative) understand the risks of neutropenia fever and its management. Patient Feedback: 92% of patients and caregivers reported increased confidence in managing fever-related emergencies due to real-time alerts.
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FINANCIAL PROGRESS OF THE PROJECT

Month	Admin Charge	Devices	Baseline, awareness sessions, promotional activity, tie up with hospitals and deployment with follow-ups, travel & subsistence
October	12500.00	500000.00	40000.00
November	12500.00	0.00	60000.00
December	12500.00	0.00	100000.00
January	12500.00	0.00	100000.00

February	12500.00	500000.00	100000.00
March	12500.00	0.00	25000.00
Total	75000	1000000	425000

IMPACT & SUSTAINABILITY OF THE NEUTROPENIA FEVER AWARENESS & MONITORING PROJECT

Impact Area	Achievements
Early Fever Detection	Multiple cases of neutropenia fever were detected early, allowing timely medical intervention.
Improved Patient Outcomes	Many patients who received timely intervention experienced better recovery and reduced complications.
Detection of Other Fevers	The program also identified cases of viral and bacterial infections, prompting necessary medical attention.
Increased Awareness	A significant number of patients & caregivers attended awareness sessions, improving their understanding of fever risks.

SUSTAINABILITY OF THE PROGRAM

Sustainability Strategy	Implementation Approach
Device Reuse & Rotation	Hospital staff trained to efficiently redistribute monitoring devices to new patients, ensuring long-term usage.
Capacity Building	Healthcare workers and volunteers equipped with skills to integrate early fever detection
Community Empowerment	Caregivers educated on symptom recognition and encouraged to act promptly on warning signs.
Partnerships for Scale-up	Collaboration with NGOs, CSR partners, and healthcare providers to extend the program's reach.

ADDRESSING POOR INFRASTRUCTURE & SOCIAL CHALLENGES

- Limited Healthcare Access - Strengthened teleconsultation and referral networks to connect patients with doctors remotely.
- Low Awareness & Health Literacy - Continued community-based education and IEC materials in local languages to bridge knowledge gaps.